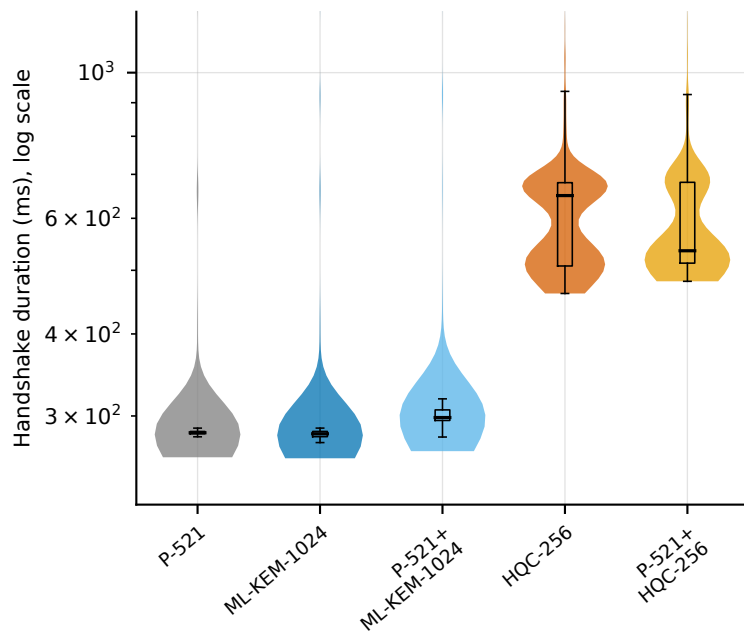
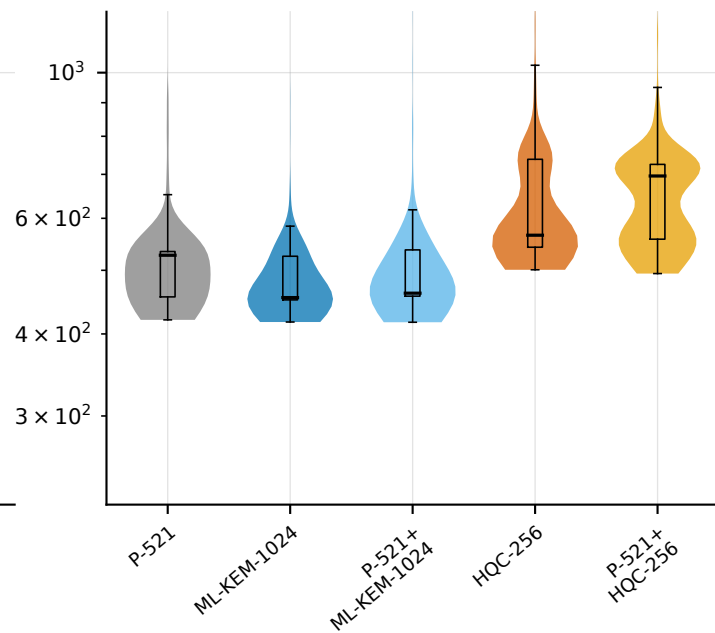


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	299.2	37.9	1.33k	521.8	28.7	1.62k
ML-KEM-1024	298.3	38.3	1.34k	488.5	26.3	1.58k
P-521+ML-KEM-1024	320.5	42.5	1.45k	509.4	32.2	1.79k
HQC-256	611.8	21.1	1.69k	638.6	29.3	2.61k
P-521+HQC-256	598.4	23.4	1.71k	665.5	24.6	1.78k